From EEG Intent to Simulated Robotic Pickup: A Midterm Project Reflection

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Abstract

This project asks whether a low-bandwidth motor-imagery EEG interface can reliably enable, pause, and resume a learned simulated robot pickup skill. The original gap was the missing evaluation between offline decoding and physical task behavior: a plausible classifier score does not show that noisy commands produce a useful grasp. We separated calibrated rest/hand-imagery decoding, causal replay and command gating, and a learned manipulation policy. A corrected five-person scripted comparison and one exploratory EEG-to-learned-policy replay now exist; the latter retained a lift with raw commands in three development runs, whereas confidence and temporal gates produced no acquisitions. The remaining work is a frozen held-out comparison that measures completion together with unintended rest motion and missed imagery, without treating replay as live BCI validation.

1 Introduction and objective

Non-invasive EEG may provide an assistive command channel, but motor-imagery signals are noisy and differ across people and sessions [4, 5]. An offline rest-versus-movement score also leaves an important systems question unanswered: can the resulting command stream control a robot task within a fixed time budget? Benchmarking work has shown why results from one dataset or split should not be assumed to generalize [3]. The original proposal therefore targeted a reproducible, simulation-first bridge from public EEG to robotic action rather than a live or clinical device.

The key insight was to divide responsibilities. A decoder estimates only a small vocabulary from protocol-defined EEG labels; a gate turns changing estimates into enable or HOLD requests; a learned robot skill handles continuous joint motion. This follows the broader shared-autonomy idea of giving the user a high-level choice while the robot executes lower-level behavior [1]. In our PhysioNet recordings, the usable labels are instructed rest and left/right fist imagery, combined as hand imagery [6]. Mapping imagery to a pickup request is our interface choice; it is not evidence that the dataset labels grasp intent.

The original twelve-week plan called for leakage-aware EEG baselines, optional Riemannian and EEGNet comparisons, causal replay, temporal filtering, Isaac Lab policy learning, then matched end-to-end tests. The intended outcome was a measurable simulated hand task, with decoder errors, command stability, and physical completion reported separately. Figure 1 shows the implemented boundary and the remaining evaluation step.

2 Reflection: where are we?

What worked. We implemented a reproducible CSP/LDA decoder and causal command replay. On five later participants (S006–S010), using each person's runs 4 and 8 for calibration and run 12 for evaluation,

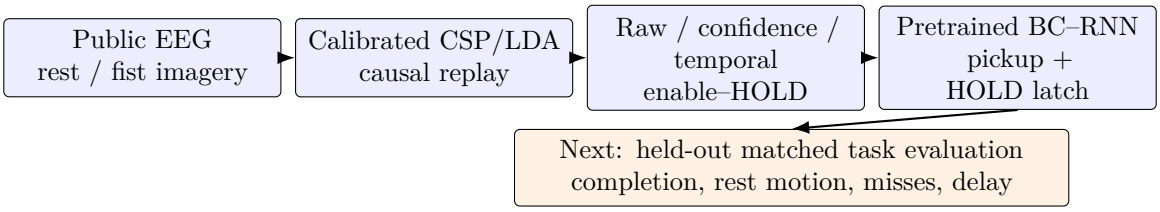

Figure 1: Project overview. Blue stages have executable components; the orange evaluation is planned. The EEG labels specify instructed rest or fist imagery, not a grasp. The robot skill is an externally pretrained policy with a project-built HOLD/completion adapter. The one-subject learned-policy result is exploratory; the held-out comparison has not been run.

balanced accuracy was 59.33% across 150 windows; rest-window false activation was 29.33%. This is modest evidence that the data contain decodable information, not proof of reliable online control. A corrected 120-second, five-person *scripted* robot comparison passed its HOLD, uninterrupted, and pause/resume controls. Raw, confidence, and temporal gates yielded physical lifts in 5/5, 3/5, and 2/5 cases. Mean command activation during annotated rest fell from 38.7% to 25.7% and 19.6%, respectively, while missed imagery bouts rose from 12 to 30 and 38. Thus the predicted command-stability benefit is real in this fixed replay, but it comes with a task-completion cost.

Robot learning advanced unevenly. The original Isaac Lab PPO policy lifted from privileged in-hand starts but did not reliably pick up a cube from a table. A later fixed near-table policy reached 2/5 sampled acquisitions, with a drop and large asymmetric finger forces; it was not a qualified pickup controller. We therefore tested an attributed, externally pretrained robomimic BC–RNN policy in its compatible native simulator. Its first five rollouts acquired and lifted but retained the cube in 0/5 terminal states. A project-built continuous enable/HOLD/resume adapter and completion latch addressed that measured retention failure. In a later 24-run development control batch, 21/21 positive conditions acquired and retained at the terminal state, while 0/3 always-HOLD negatives acquired. Those runs reused development seeds and do not estimate generalization.

The first EEG-to-*learned*-policy replay is now complete for S006 run 12. With an identical pretrained checkpoint and three development scene seeds, raw commands acquired and retained in 3/3 runs; confidence and temporal commands acquired in 0/3 because they enabled the policy for only 24 and 14 of 400 steps, respectively. This verifies a narrow version of the original insight: the high-level interface can drive a learned simulated skill, and gating materially changes physical outcomes. It does *not* verify the stronger hypothesis that temporal filtering improves task completion, nor does it establish held-out, live-user, cross-person, or hardware performance. S006 was a decoder development/training subject, and this experiment replayed a saved decision trace rather than rerunning a live decoder.

What was harder, and was the gap real? The gap between offline decoding and useful action proved concrete: a 59.33% balanced decoder can produce many rest false activations, and filters that suppress them can withhold the sustained enable time needed for pickup. The original plan underestimated robot acquisition and retention, HOLD mechanics, and the importance of measuring contact and terminal cube height rather than trusting reward or a brief lift. The gap is a practical evaluation gap in this project; the literature already contains shared-autonomy and pseudo-online BCI methods, so we do not claim a new algorithm or that no one has studied the bridge before [1, 2].

3 Updated work plan

The revised deliverable is a bounded shared-autonomy systems study using the existing binary EEG vocabulary and an attributed learned policy. The scripted robot remains a diagnostic reference; the

learned-policy comparison is the main result. We will report both the cost and benefit of each gate, even if raw commands complete more tasks.

Original item	Status	Change and reason
EEG baseline and replay	Done	Keep calibrated binary labels; report the observed false activations and misses.
Riemannian/EEGNet and fine-grained gestures	Cut	Available labels do not identify wrist, twist, or grasp; more models would not fix the measured interface bottleneck.
Isaac PPO table pickup	In progress	Preserve negative and partial results; use the qualified native pretrained skill for the end-to-end study.
EEG-gate to learned robot	In progress	S006 development replay works for raw commands; test new people and matched controls next.
HOLD and retention checks	Added	Brief lifts and frozen commands did not guarantee terminal retention; the adapter and latch need independent checks.

In the week of November 9, freeze the held-out protocol, scene schedule, checkpoint, gate parameters, and physical success criteria before examining results. In the weeks of November 16 and 23, run S001–S003 as a held-out exploratory cohort with matched raw, confidence, temporal, always-HOLD, always-enable, and shuffled-command conditions; retain failed and incomplete trials. In the week of November 30, analyze per-person completion, rest motion, missed bouts, delays, and drops, and audit trace provenance. In the week of December 7, package the report, figure, reproduction instructions, and limitations. If held-out pickup or filtering fails, the viable final result is the measured failure and its mechanism, supported by the complete scripted comparison and the S006 development demonstration; no success threshold will be relaxed after seeing the held-out cases. Broader scene variation, live EEG, Tesollo hardware, and extra EEG model families are deferred.

4 AI statement

AI tools assisted with literature organization, code and experiment workflow, result summaries, and editing this report. I checked numerical claims against saved project records and distinguished the externally pretrained policy from project-built code. The project conclusions and stated limitations remain my responsibility.

References

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